

Problems with Neural Networks

Vanishing & Exploding gradients

- weight init
- Activation function
- Batch Normalization
- Gradient Clipping

Not enough data

- transfer learning
- Unsupervised pre-training

Slow training

- Optimizers (Adam)
- Learning rate scheduler

Overfitting

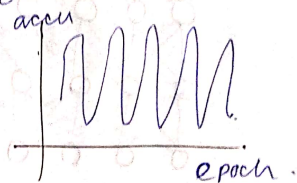
- L_1, L_2 Regularization
- Dropouts

Different range values of features (inputs)

While using unscaled data in ANNs →

there most probably the algorithm cannot converge to the best solution.

you can see an oscillatory motion in the accuracy curve

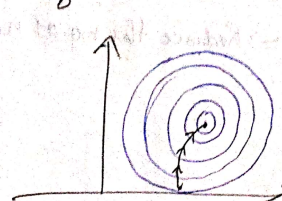
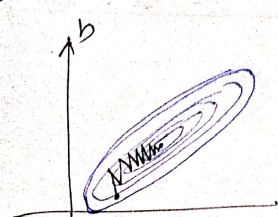
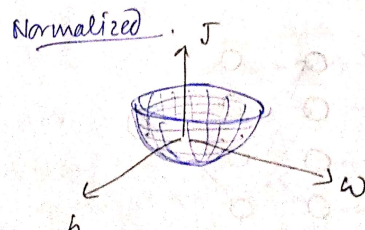
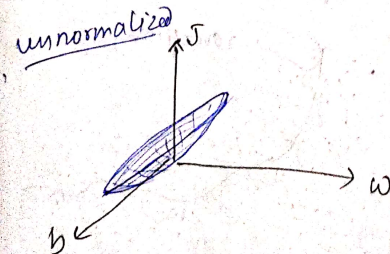


age range, range, range
 x_1, x_2 (Salary)

$$w_1, w_2 - \eta \frac{\partial L}{\partial w}$$

focus on this weight more because x_2 is more focused.

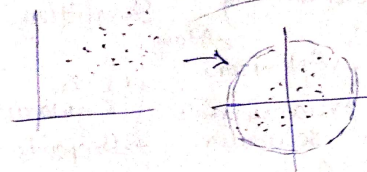
All features brought to a comparable scale, GD can update all weights fairly



When we don't know of min/max

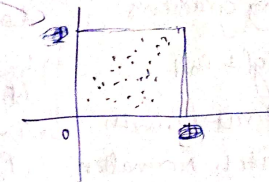
Standardize

$$\frac{x_i - \mu}{\sigma}$$



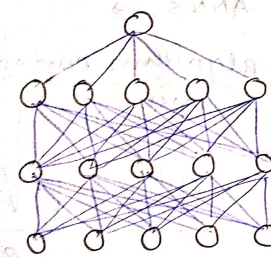
Normalize

$$\frac{x_i - x_{min}}{x_{max} - x_{min}}$$

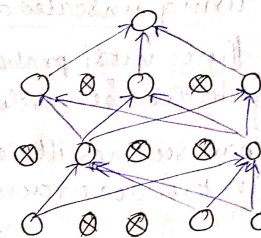


When we know of min-max

The concept of dropouts



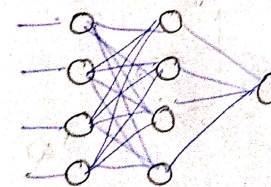
Standard NN



Applying dropout

randomly drop some neurons in each epoch.

Actually training a fixed data but training with different NNs in each epoch.



→ Reduce the no. of nodes

